UNIVERSITÀ DI PARMA

ANSI C99

Informatica e Laboratorio di Programmazione

The true sign of intelligence is not knowledge but imagination.

Albert Einstein

Lessons

- ~ ⅔ in Class (slides + code examples)
- ~ ⅓ in Laboratory (actual programming)
- Slides, code examples, and laboratory assignments
  - https://elly2025.didattica.unipr.it
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- Candidates should
  - know basic concepts of Information Technology
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Course program

- Theory
  - Class
- Practical part AKA programming
  - Class + Laboratory
- Problem solving
  - Class + Laboratory

Course organization

- Theory (Class)
- Basic information needed to understand how to program
  - Basic computer architecture
  - Algorithms
  - How numbers can be managed in a computer

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- Programming part (class + laboratory)
  - Centered on the programming language and algorithms
  - C fundamentals
  - Variables
  - Control instructions
  - Expressions
  - Array, strings & pointers
  - Functions and recursion
  - I/O
- Slides + code examples

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- You **must** use your official UNIPR account:
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- You will be given more detailed instructions later
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  - C
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Exam results

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Correzioni online di Informatica & Laboratorio di Programmazione

Menù

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- Do not underestimate the effort needed to pass the exam
- Usually students that do not attend lessons $\longrightarrow$ negative result
- Some statistics (2023 cohort)
  - Passed: 52.2% (48 students out of 102)
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  - Time average: 1.9 months
  - Average n. of attempts: 1.8
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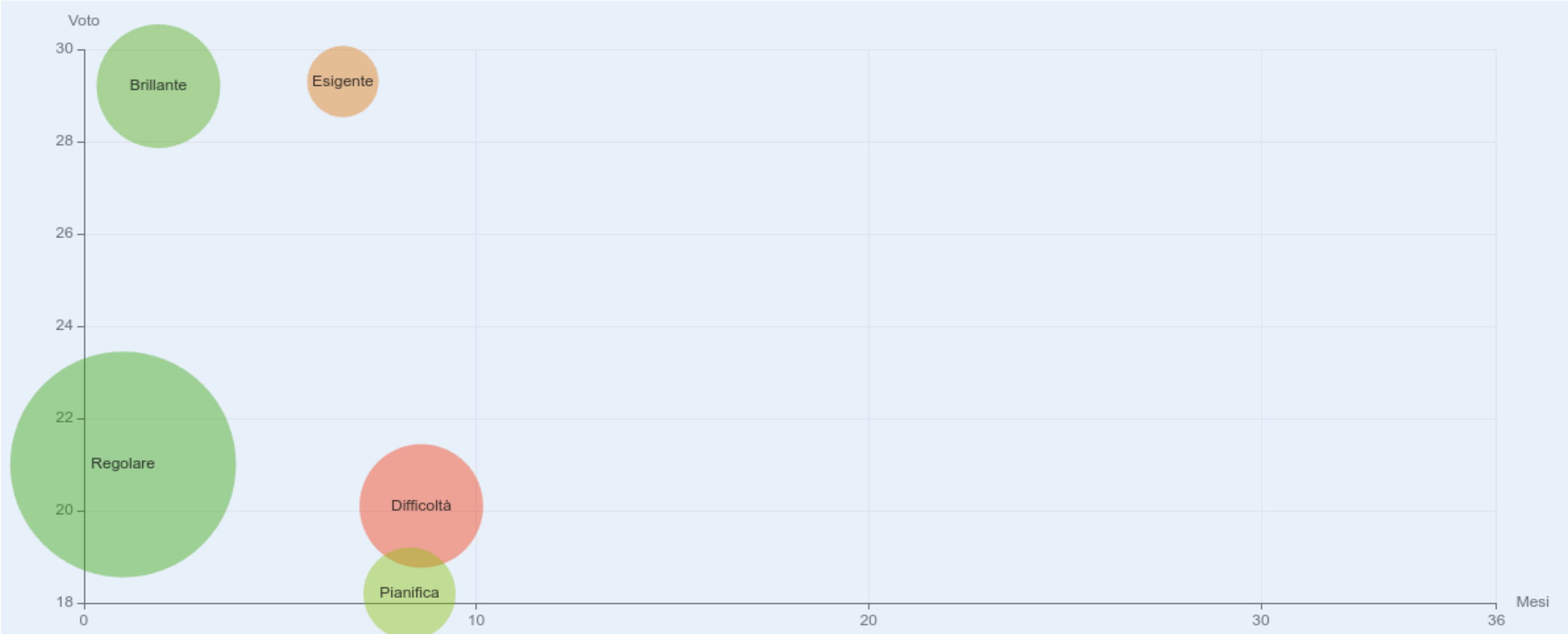

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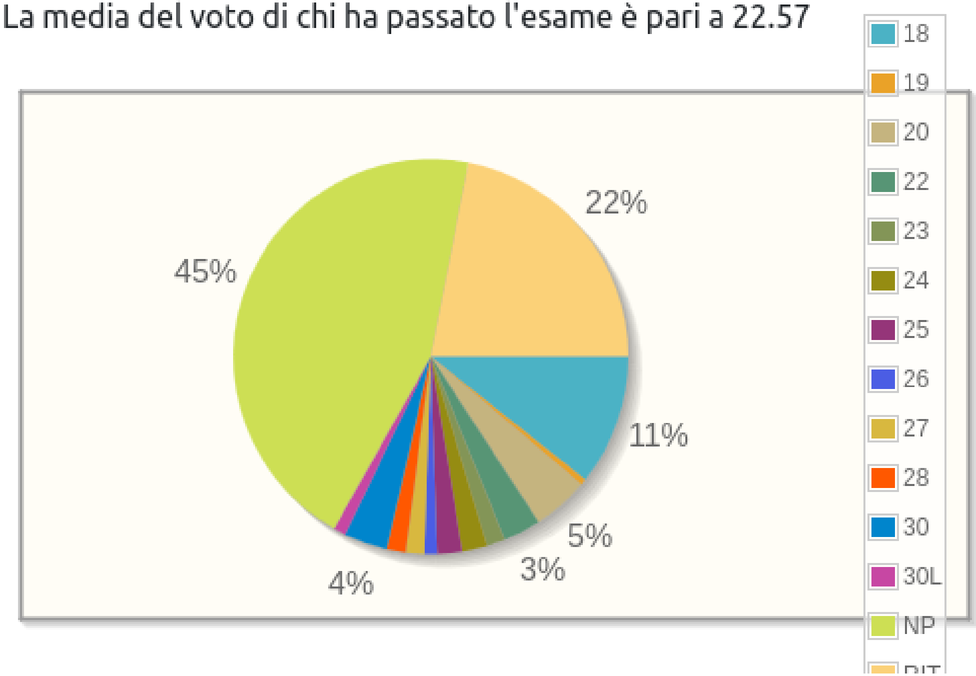

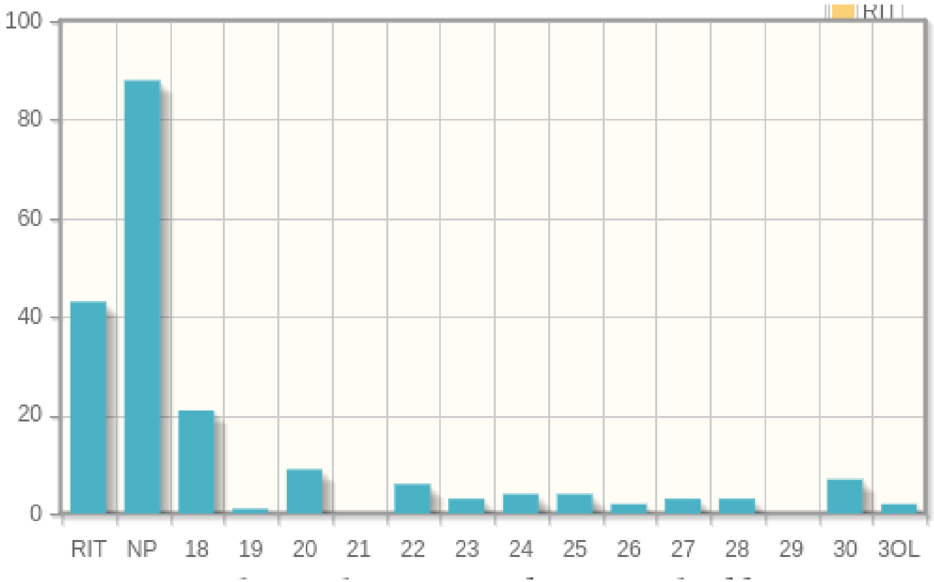

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  - Any book that can be a good **reference** text

C only

Also Programming

Textbooks

Returns `c` (the character written). If a write error occurs, `putchar` sets the stream's error indicator and returns `EOF`. *7.3, 22.4*

`puts` *Write String* `<stdio.h>`

```
int puts(const char *s);
```

Writes the string pointed to by `s` to the `stdout` stream, then writes a new-line character.

Returns A nonnegative value if successful. Returns `EOF` if a write error occurs. *13.3, 22.5*

`putwc` *Write Wide Character to File (C99)* `<wchar.h>`

```
wint_t putwc(wchar_t c, FILE *stream);
```

Wide-character version of `putc`. *25.5*

`putwchar` *Write Wide Character (C99)* `<wchar.h>`

```
wint_t putwchar(wchar_t c);
```

Wide-character version of `putchar`. *25.5*

`qsort` *Sort Array* `<stdlib.h>`

```
void qsort(void *base, size_t nmemb, size_t size,
           int (*compar)(const void *, const void *));
```

Sorts the array pointed to by `base`. The array has `nmemb` elements, each `size` bytes long. `compar` is a pointer to a comparison function. When passed pointers to two array elements, the comparison function must return a negative, zero, or positive integer, depending on whether the first array element is less than, equal to, or greater than the second. *17.7, 26.2*

`raise` *Raise Signal* `<signal.h>`

```
int raise(int sig);
```

Raises the signal whose number is `sig`.

Returns Zero if successful, nonzero otherwise. *24.3*

`rand` *Generate Pseudo-Random Number* `<stdlib.h>`

```
int rand(void);
```

Returns A pseudo-random integer between 0 and `RAND_MAX` (inclusive). *26.2*

`realloc` *Resize Memory Block* `<stdlib.h>`

```
void *realloc(void *ptr, size_t size);
```

`ptr` is assumed to point to a block of memory previously obtained from `calloc`, `malloc`, or `realloc`. `realloc` allocates a block of `size` bytes, copying the contents of the old block if necessary.

Returns A pointer to the beginning of the new memory block. Returns a null pointer if a block of the requested size can't be allocated. *17.3*

Textbooks

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A.14.2 Pseudo-Random Number Generator Functions

The *rand()* and *srand()* functions enable you to generate pseudo-random numbers.

A.14.2.1 The *rand()* Function

```
#include <stdlib.h>
int rand( void );
```

The *rand()* function returns an integer in the range 0 through RAND_MAX. Successive calls to *rand()* should produce different integers. However, the sequence of random numbers could be the same for each program execution unless you use a different seed value via the *srand()* function.

A.14.2.2 The *srand()* Function

```
#include <stdlib.h>
void srand( unsigned int seed );
```

The *srand()* function uses the argument as a seed for a new sequence of pseudo-random numbers to be returned by subsequent calls to *rand()*. If *srand()* is invoked with the same seed value, the sequence of generated numbers will be the same. The default seed value is 1.

A.14.3 Memory Management Functions

The memory management functions enable you to allocate and deallocate memory dynamically. See Chapter 7 for more information about these functions.

A.14.3.1 The *calloc()* Function

```
#include <stdlib.h>
void *calloc( size_t nmemb, size_t size );
```

The *calloc()* function allocates contiguous space for *nmemb* objects, each of which has a length in bytes specified by *size*. All bits in the allocated space are initialized to zero. *calloc()* returns a pointer to the first byte of the allocated space. If the space cannot be allocated, or if *nelem* or *size* is zero, *calloc()* returns a null pointer.

Tips

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- **Be curious:** you are here to learn, not to pass the exam
  - If you learn you are going to pass the exam anyway
- **Ask questions:** do not be shy, no question is stupid
  - If something is not clear, stop me as soon as possible
- **Be active:** learning by doing is the key for success
  - Exercise your self

Contacts

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- Massimo Bertozzi
- bertozzi@ce.unipr.it
- 0521 90 5845

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Informatica e Laboratorio di Programmazione

nec scire fas est omnia

Quinto Orazio Flacco, Odi

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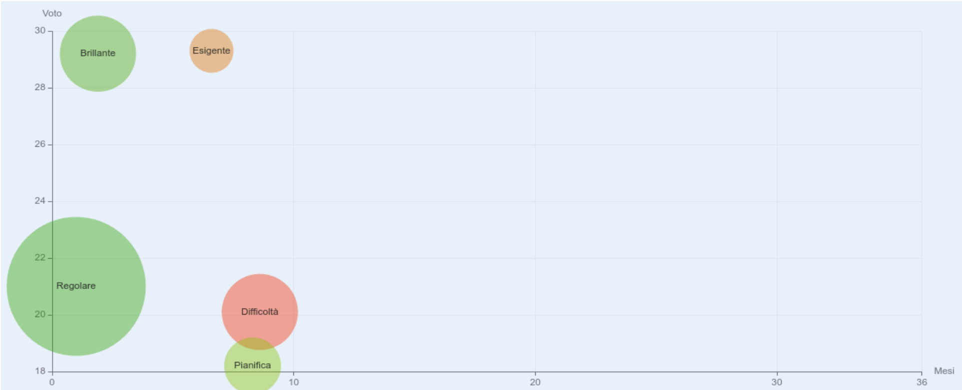

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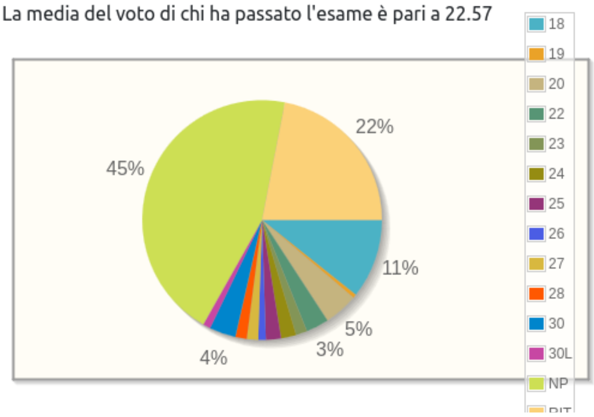

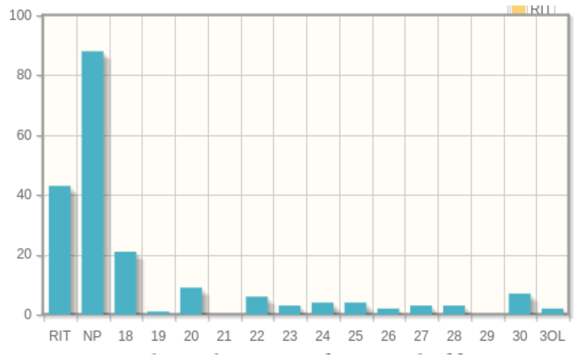

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